Physical Chemistry (I)

Lecture 10

Helmholtz and Gibbs Energies

Where Are We?

Foundations

- Basic concepts
- Gases
- Thermodynamics
- Statistical mechanics

Applications

- Phases
- Simple mixtures
- Chemical equilibrium
- Electrochemistry

“Chapters” in This Course

1. Basic concepts (1, 2)
2. Gases (2)
3. The first law (1)
4. Entropy (3)
5. Helmholtz and Gibbs energies (3)
6. Statistical mechanics (X)
7. Phase transitions (4)
8. Simple mixtures (5)
9. Chemical equilibria (6)
10. Electrochemistry (6)

5. Helmholtz and Gibbs Energies

Chapter Overview

- Helmholtz and Gibbs energies
  - Definitions
  - Meanings
- Thermodynamics potentials
  - Differential forms
  - Properties of exact differentials
- Thermochemistry of Gibbs energy

Clausius Inequality

$$\mathrm{d}S \geq \frac{\delta q}{T}$$

$$\mathrm{d}S - \frac{\delta q}{T} \geq 0$$

1. Constant V:

$$\mathrm{d}S - \frac{\mathrm{d}U}{T} \geq 0$$

2. Constant p:

$$\mathrm{d}S - \frac{\mathrm{d}H}{T} \geq 0$$

Spontaneity

1. Constant V:

$$\mathrm{d}S - \frac{\mathrm{d}U}{T} \geq 0$$

$$\mathrm{d}(\underbrace{U - TS}_{\text{Helmholtz energy } A}) \leq 0 \text{ (at constant } T\text{)}$$

2. Constant p:

$$\mathrm{d}S - \frac{\mathrm{d}H}{T} \geq 0$$

$$\mathrm{d}(\underbrace{H - TS}_{\text{Gibbs energy } G}) \leq 0 \text{ (at constant } T\text{)}$$

Helmholtz and Gibbs Energies

Helmholtz energy $A = U - TS$

Gibbs energy $G = H - TS$

Criteria of spontaneity:

$$\mathrm{d}A_{T,V} \leq 0$$

(At constant T and V, $\mathrm{d}A \propto -\mathrm{d}S_{\mathrm{total}}$)

$$\mathrm{d}G_{T,p} \leq 0$$

(At constant T and p, $\mathrm{d}G \propto -\mathrm{d}S_{\mathrm{total}}$)

So, What Do They Mean?

- Helmholtz energy = maximum work
- Gibbs energy = maximum non-expansion work

A and Maximum Work

$$T\mathrm{d}S \geq \delta q \text{ (Clausius inequality)}$$

$$\mathrm{d}U = \delta q + \delta w \text{ (first law)}$$

Hence,

$$T\mathrm{d}S \geq \mathrm{d}U - \delta w$$

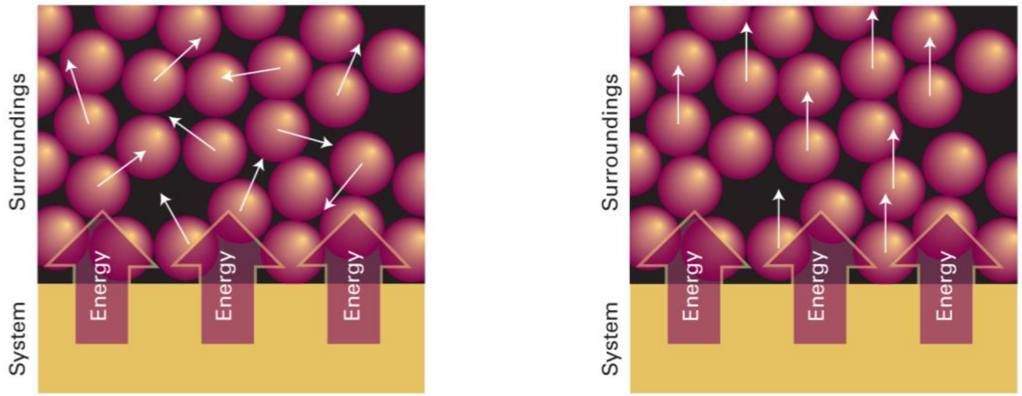
$$\delta w \geq \mathrm{d}U - T\mathrm{d}S = \mathrm{d}A_T$$

As $\delta w < 0$ for the work by the system,

$$\mathrm{d}A_T = \delta w_{\mathrm{max}}$$

$$\Delta A_T = w_{\mathrm{max}}$$

Meaning of Helmholtz Energy



$$A = U - TS$$

At temperature T,

TS = energy stored as thermal motion

$U - TS$ = maximum work obtainable from the system

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figures 2A.3 and 2A.4)

G and Maximum Non-Exp Work

$$H = U + pV$$

$$\mathrm{d}H = \mathrm{d}U + \mathrm{d}(pV)$$

$$\mathrm{d}H = \delta q + \delta w + \mathrm{d}(pV)$$ (first law)

Recall $G = H - TS$. At constant T,

$$\mathrm{d}G_T = \mathrm{d}H - T\mathrm{d}S = \delta q + \delta w + \mathrm{d}(pV) - T\mathrm{d}S$$

For a reversible change, $\delta q_{\mathrm{rev}} = T\mathrm{d}S$

$$\mathrm{d}G_T = \delta w_{\mathrm{rev}} + \mathrm{d}(pV)$$

G and Maximum Non-Exp Work

$$\mathrm{d}G_T = \delta w_{\mathrm{rev}} + \mathrm{d}(pV)$$

At constant p,

$$\mathrm{d}G_{T,p} = \delta w_{\mathrm{rev}} + p\mathrm{d}V$$

On the other hand,

$$\delta w_{\mathrm{rev}} = -p\mathrm{d}V + \delta w_{\mathrm{add,rev}}$$

Thus,

$$\mathrm{d}G_{T,p} = \delta w_{\mathrm{add,rev}} = \delta w_{\mathrm{add,max}}$$

$$\Delta G_{T,p} = w_{\mathrm{add,rev}} = w_{\mathrm{add,max}}$$

Thermodynamic Potentials

$+pV$

$-TS$

U	$H = U + pV$
$A = U - TS$	$G = U + pV - TS$

Differential Forms

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$ (fundamental equation)

- Enthalpy $H = U + pV$

$$\mathrm{d}H = \mathrm{d}U + p\mathrm{d}V + V\mathrm{d}p = (T\mathrm{d}S - \cancel{p\mathrm{d}V}) + \cancel{p\mathrm{d}V} + V\mathrm{d}p$$

$$\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$$

- Helmholtz energy $A = U - TS$

$$\mathrm{d}A = \mathrm{d}U - T\mathrm{d}S - S\mathrm{d}T = (\cancel{T\mathrm{d}S} - p\mathrm{d}V) - \cancel{T\mathrm{d}S} - S\mathrm{d}T$$

$$\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V$$

Differential Forms

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$ (fundamental equation)

- Gibbs energy $G = U + pV - TS$

$$\mathrm{d}G = \mathrm{d}U + p\mathrm{d}V + V\mathrm{d}p - T\mathrm{d}S - S\mathrm{d}T$$

$$\mathrm{d}G = (\cancel{T\mathrm{d}S} - \cancel{p\mathrm{d}V}) + \cancel{p\mathrm{d}V} + V\mathrm{d}p - \cancel{T\mathrm{d}S} - S\mathrm{d}T$$

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p$$

Natural Independent Variables

- Internal energy $U = U(S, V)$

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$

- Enthalpy $H = U + pV = H(S, p)$

$$\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$$

- Helmholtz energy $A = U - TS = A(T, V)$

$$\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V$$

- Gibbs energy $G = U + pV - TS = G(T, p)$

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p$$

Not in Atkins

Exact Differentials

The thermodynamic potentials are state functions

For $H = H(S, p)$,

$$\mathrm{d}H = \left(\frac{\partial H}{\partial S}\right)_p \mathrm{d}S + \left(\frac{\partial H}{\partial p}\right)_S \mathrm{d}p$$

Comparing this to $\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$,

$$\left(\frac{\partial H}{\partial S}\right)_p = T, \qquad \left(\frac{\partial H}{\partial p}\right)_S = V$$

Not in Atkins

Exact Differentials

Similarly, using $A = A(T, V)$, one can show

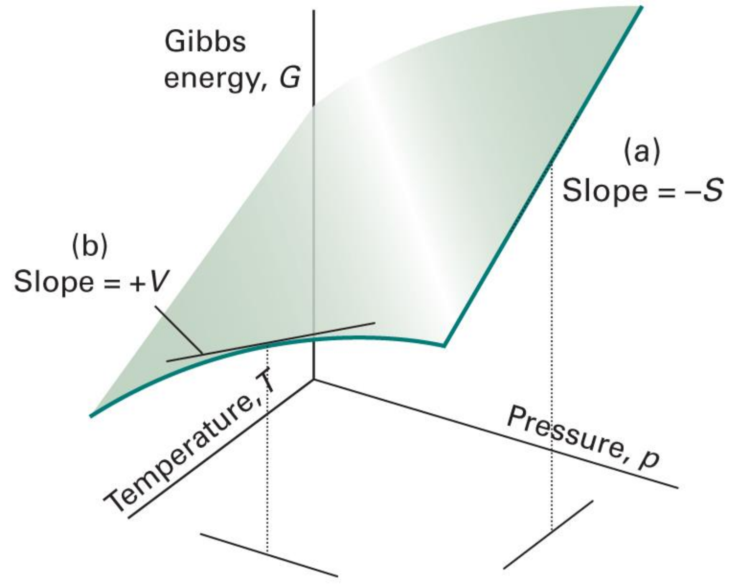
$$\left(\frac{\partial A}{\partial T}\right)_V = -S, \qquad \left(\frac{\partial A}{\partial V}\right)_T = -p$$

Using $G = G(T, p)$, one can show

$$\left(\frac{\partial G}{\partial T}\right)_p = -S, \qquad \left(\frac{\partial G}{\partial p}\right)_T = V$$

Example: Gibbs Energy

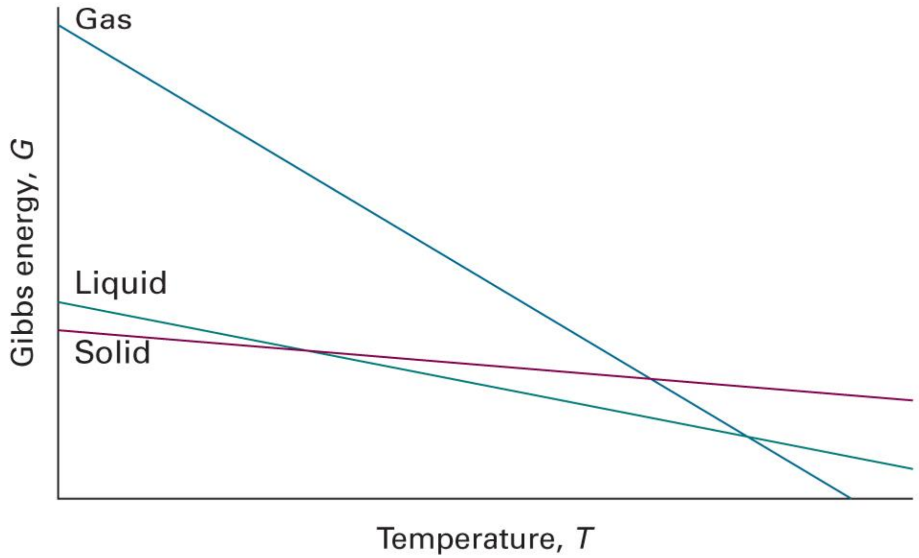
$$\left(\frac{\partial G}{\partial T}\right)_p = -S, \qquad \left(\frac{\partial G}{\partial p}\right)_T = V$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figures 3E.1)

Physical Interpretation

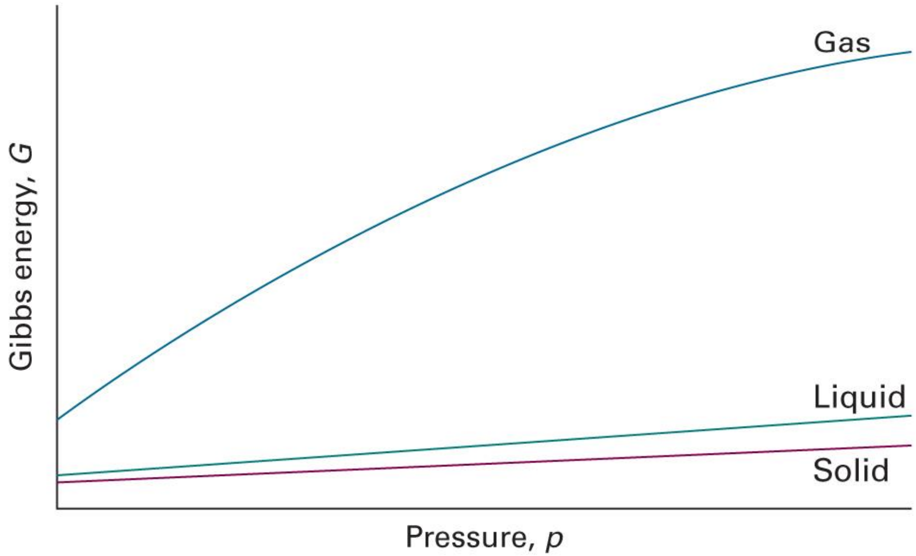
$$\left(\frac{\partial G}{\partial T}\right)_p = -S < 0$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figures 3E.2)

Physical Interpretation

$$\left(\frac{\partial G}{\partial p}\right)_T = V > 0$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figures 3E.3)

$\boldsymbol{G}$ and $\boldsymbol{T}$

$$\left(\frac{\partial G}{\partial T}\right)_p = -S$$

Since $G = H - TS$,

$$\left(\frac{\partial G}{\partial T}\right)_p = \frac{G - H}{T}$$

Topic 3E

Gibbs-Helmholtz Equation

$$\left(\frac{\partial(G/T)}{\partial T}\right)_p = \left[\frac{\partial}{\partial T}\left(G \times \frac{1}{T}\right)\right]_p = \frac{1}{T}\left(\frac{\partial G}{\partial T}\right)_p + G\left[\frac{\partial}{\partial T}\left(\frac{1}{T}\right)\right]_p$$

$$\left(\frac{\partial(G/T)}{\partial T}\right)_p = \frac{1}{T}\left(\frac{\partial G}{\partial T}\right)_p - \frac{G}{T^2}$$

Since $(\partial G/\partial T)_p = (G - H)/T$,

$$\left(\frac{\partial(G/T)}{\partial T}\right)_p = \frac{G-H}{T^2} - \frac{G}{T^2} = -\frac{H}{T^2}$$

Topic 3E

Application of G-H Equation

$$\left(\frac{\partial(G/T)}{\partial T}\right)_p = -\frac{H}{T^2}$$

At constant T,

$$\left(\frac{\partial(\Delta G/T)}{\partial T}\right)_p = -\frac{\Delta H}{T^2}$$

Property 1 of Exact Differential

If f is a function of x and y, the differential $\mathrm{d}f$ is

$$\mathrm{d}f = \left(\frac{\partial f}{\partial x}\right)_y \mathrm{d}x + \left(\frac{\partial f}{\partial y}\right)_x \mathrm{d}y$$

when $\mathrm{d}f$ is an exact differential.

Then, we have

$$\left(\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)_y\right)_x = \left(\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)_x\right)_y$$

(Schwartz's theorem)

Maxwell Relations

For an exact differential $\mathrm{d}f = g\mathrm{d}x + h\mathrm{d}y$,

$$\left(\frac{\partial g}{\partial y}\right)_x = \left(\frac{\partial h}{\partial x}\right)_y$$

- Internal energy $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$$

- Enthalpy $\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$

Maxwell Relations

For an exact differential $\mathrm{d}f = g\mathrm{d}x + h\mathrm{d}y$,

$$\left(\frac{\partial g}{\partial y}\right)_x = \left(\frac{\partial h}{\partial x}\right)_y$$

- Helmholtz energy $\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

- Gibbs energy $\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p$

$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$$

Example 3E.1

Use the Maxwell relations to show that the entropy of a perfect gas is linearly dependent on $\ln V$, that is, $S = a + b \ln V$.

From the Maxwell relations,

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

For a perfect gas,

$$\left(\frac{\partial p}{\partial T}\right)_V = \left[\frac{\partial}{\partial T}\left(\frac{nRT}{V}\right)\right]_V = \frac{nR}{V}$$

Hence,

$$\left(\frac{\partial S}{\partial V}\right)_T = \frac{nR}{V}$$

Example 3E.1

$$\left(\frac{\partial S}{\partial V}\right)_T = \frac{nR}{V}$$

At constant temperature,

$$\mathrm{d}S = \frac{nR}{V}\mathrm{d}V$$

$$\int \mathrm{d}S = \int \frac{nR}{V}\mathrm{d}V$$

Therefore,

$$S = nR\ln V + C$$

Property 2 of Exact Differential

If f is a function of x and y, the differential $\mathrm{d}f$ is

$$\mathrm{d}f = \left(\frac{\partial f}{\partial x}\right)_y \mathrm{d}x + \left(\frac{\partial f}{\partial y}\right)_x \mathrm{d}y$$

when $\mathrm{d}f$ is an exact differential.

Differentiating with respect to x at constant z gives

$$\left(\frac{\partial f}{\partial x}\right)_z = \left(\frac{\partial f}{\partial x}\right)_y + \left(\frac{\partial f}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_z$$

Internal Pressure

Internal pressure $\pi_T = (\partial U/\partial V)_T$

From property 2,

$$\left(\frac{\partial U}{\partial V}\right)_T = \left(\frac{\partial U}{\partial V}\right)_S + \left(\frac{\partial U}{\partial S}\right)_V \left(\frac{\partial S}{\partial V}\right)_T$$

Since $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$,

$$\pi_T = -p + T\left(\frac{\partial S}{\partial V}\right)_T$$

Thermodynamic Eq. of State

$$\pi_T = T\left(\frac{\partial S}{\partial V}\right)_T - p$$

From the Maxwell relation $(\partial S/\partial V)_T = (\partial p/\partial T)_V$,

$$\pi_T = T\left(\frac{\partial p}{\partial T}\right)_V - p$$

or,

$$p = T\left(\frac{\partial p}{\partial T}\right)_V - \pi_T$$

(thermodynamic equation of state)

Example 3E.2

Show thermodynamically that $\pi_T = 0$ for a perfect gas, and compute its value for a van der Waals gas.

For a perfect gas,

$$\left(\frac{\partial p}{\partial T}\right)_V = \left[\frac{\partial}{\partial T}\left(\frac{nRT}{V}\right)\right]_V = \frac{nR}{V}$$

From the thermodynamic equation of state,

$$\pi_T = T\left(\frac{\partial p}{\partial T}\right)_V - p = T\left(\frac{nR}{V}\right) - p = \frac{nRT}{V} - p = p - p = 0$$

Example 3E.2

For a van der Waals gas,

$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2}$$

$$\left(\frac{\partial p}{\partial T}\right)_V = \left[\frac{\partial}{\partial T}\left(\frac{nRT}{V - nb} - a\frac{n^2}{V^2}\right)\right]_V = \frac{nR}{V - nb}$$

From the thermodynamic equation of state,

$$\pi_T = T\left(\frac{\partial p}{\partial T}\right)_V - p = T\left(\frac{nR}{V - nb}\right) - p = \frac{nRT}{V - nb} - \left(\frac{nRT}{V - nb} - a\frac{n^2}{V^2}\right)$$

$$\pi_T = a\frac{n^2}{V^2}$$

Thermochemistry of G

$$G = H - TS$$

At constant T,

$$\Delta G = \Delta H - T\Delta S$$

We can define the standard Gibbs energy of reaction:

$$\Delta_r G^\ominus = \Delta_r H^\ominus - T\Delta_r S^\ominus$$

and the standard Gibbs energy of formation:

$$\Delta_f G^\ominus = \Delta_f H^\ominus - T\Delta_f S^\ominus$$

Standard Reaction Potentials

$$\Delta_r H^{\ominus} = \sum_{\text{products } i} \nu(i)\Delta_f H^{\ominus}(i) - \sum_{\text{reactants } j} \nu(j)\Delta_f H^{\ominus}(j)$$

$$\Delta_r S^{\ominus} = \sum_{\text{products } i} \nu(i) S_m^{\ominus}(i) - \sum_{\text{reactants } j} \nu(j) S_m^{\ominus}(j)$$

$$\Delta_r G^{\ominus} = \sum_{\text{products } i} \nu(i)\Delta_f G^{\ominus}(i) - \sum_{\text{reactants } j} \nu(j)\Delta_f G^{\ominus}(j)$$

Convention Again

$$\Delta_f G^{\ominus}(\mathrm{H}^+, aq) = 0$$

e.g. (Brief Illustration 3D.2)

$$(1/2)\mathrm{H}_2(g) + (1/2)\mathrm{Cl}_2(g) \rightarrow \mathrm{H}^+(aq) + \mathrm{Cl}^-(aq)$$

$$\Delta_r G^{\ominus} = \Delta_f G^{\ominus}(\mathrm{H}^+, aq) + \Delta_f G^{\ominus}(\mathrm{Cl}^-, aq) = \Delta_f G^{\ominus}(\mathrm{Cl}^-, aq)$$

Pressure Dependence of G

$$\mathrm{d}G = V\mathrm{d}p - S\mathrm{d}T$$

At constant T,

$$\mathrm{d}G = V\mathrm{d}p$$

Integration leads to

$$\int_{\text{state } i}^{\text{state } f} \mathrm{d}G = \int_{\text{state } i}^{\text{state } f} V\,\mathrm{d}p$$

$$\Delta G = G_f - G_i = \int_{p_i}^{p_f} V\,\mathrm{d}p$$

Pressure Dependence of G

$$\Delta G = \int_{p_i}^{p_f} V \, \mathrm{d}p$$

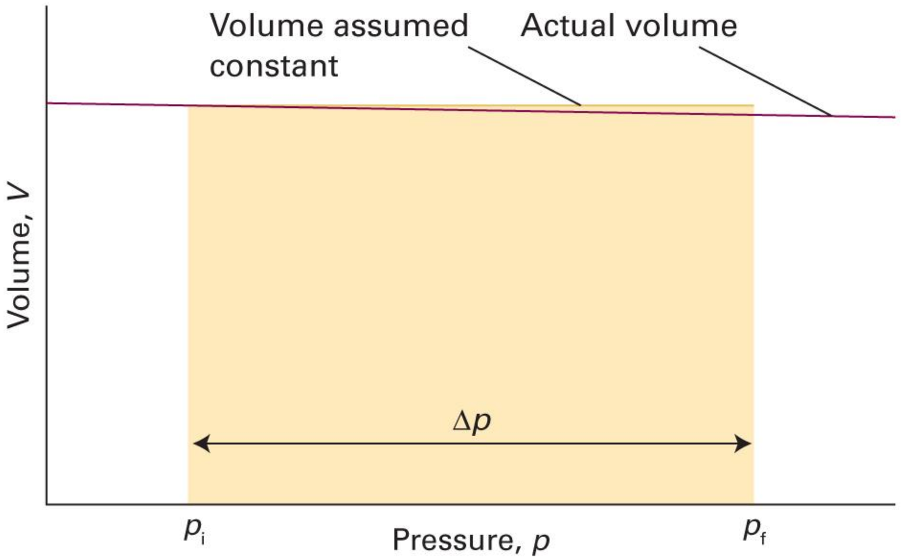
Dividing both sides by n,

$$\Delta G_{\mathrm{m}} = \int_{p_i}^{p_f} V_{\mathrm{m}} \, \mathrm{d}p$$

1. Condensed phase: $V_{\mathrm{m}} \approx$ constant
2. Gas phase: $V_{\mathrm{m}} = V/n = RT/p$

Condensed Phase

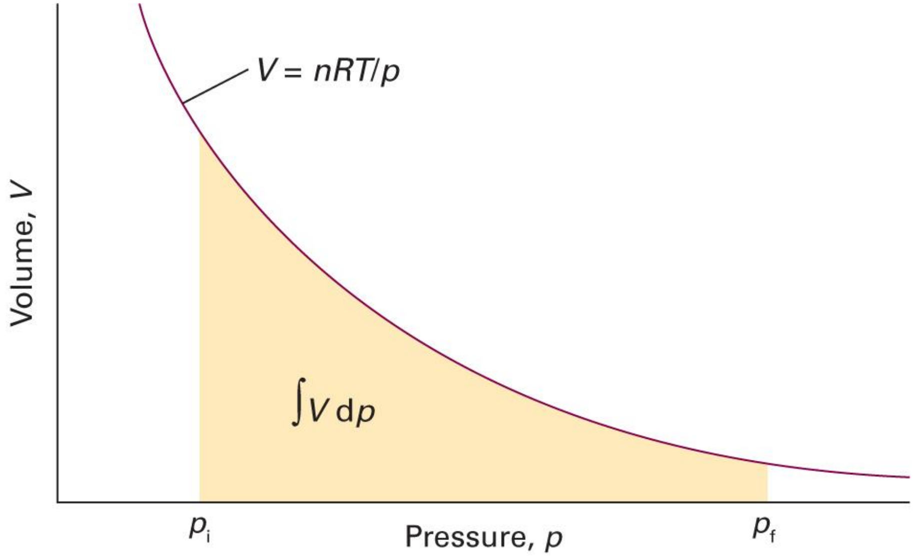
$$\Delta G_\mathrm{m} = \int_{p_i}^{p_f} V_\mathrm{m}\,\mathrm{d}p = V_\mathrm{m}\int_{p_i}^{p_f}\mathrm{d}p = V_\mathrm{m}(p_f - p_i)$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figures 3E.4)

Gas Phase

$$\Delta G_{\mathrm{m}} = \int_{p_i}^{p_f} V_{\mathrm{m}}\,\mathrm{d}p = \int_{p_i}^{p_f} \frac{RT}{p}\,\mathrm{d}p = RT\left(\ln p_f - \ln p_i\right)$$



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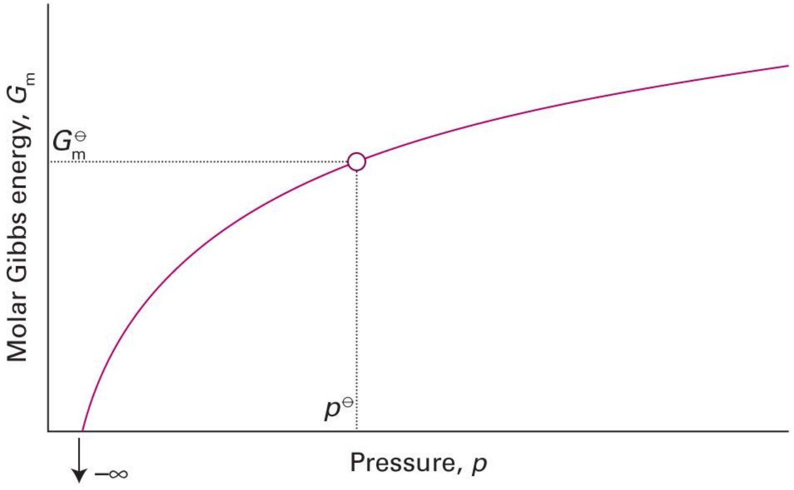
Gas Phase: Standard State

$$\Delta G_\mathrm{m} = G_\mathrm{m}(p_f) - G_\mathrm{m}(p_i) = RT\ln\frac{p_f}{p_i}$$

$$G_\mathrm{m}(p) = G_\mathrm{m}^{\ominus} + RT\ln\frac{p}{p^{\ominus}}$$



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Summary

Thermodynamic potentials

Internal energy

U

Helmholtz energy

$A = U - TS$

Enthalpy

$H = U + pV$

Gibbs energy

$G = H - TS$

Clausius ineq.

First law

Spontaneity

$\Delta A_{T,V} \leq 0$

$\Delta G_{T,p} \leq 0$

Maximum work

$A_T = w_{\max}$

$G_{T,p} = w_{\mathrm{add,max}}$

Differential forms

Thermochemistry

Applications

- Gibbs-Helmholtz equation
- Maxwell relations
- Thermodynamic equation of state

Pressure dependence